Obili Venganna Gari Sri Lekha

ms23btech11020@iith.ac.in | mysty202520@gmail.com | +91 8790153418 | LinkedIn

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.27
Raju Junior College (Allen)	Class XII (State)	2022	89.00%
Ratnam High School	Class X (State)	2020	98.00%

EXPERIENCE

Research Intern, University of New South Wales (UNSW), Sydney

- Developed a patented process for selective extraction and purification of copper from complex electronic assemblies, combining materials processing, experimentation and statistical analysis.
- Designed an automated device-depopulation platform for recovery of electronic-grade materials from end-of-life electronics and optimized processing workflows for efficiency and material quality.
- Synthesized copper nanoparticles and investigated potential applications in semiconductor fabrication and advanced packaging technologies.

UG Researcher, IIT Hyderabad

- Developed thin-film semiconductor structures using DC/RF sputtering, thermal evaporation and e-beam evaporation for device-fabrication research
- Executed cleanroom workflows including CVD, photolithography, deposition and etching; characterized materials/devices using GI-XRD, SEM and SMU.
- Used data-driven experimentation, characterization and failure analysis to optimize fabrication conditions, film quality and process reproducibility.

PROJECTS

AI Product Development / Founder -- Aidlyy AI

- Developing AI-assisted industrial systems for workflow automation and process intelligence, including an AI wearable-glasses concept and integrated AI camera module.
- Translated technical concepts into an early product proposition and secured 40+ pre-orders within 2 months, providing early market validation.
- Worked across hardware-software integration, problem framing, prototyping and iterative product development.

SENSE-STEER-CHARGE: Intelligent Reconfigurable Wireless Power Transfer Platform

- Proposed an intelligent EV wireless-charging sheet that evolves DNP's thin sheet-coil platform into a multifunctional surface integrating sensing, localization, adaptive coil activation, foreign-object detection, battery-aware power management and predictive maintenance.
- Designed a segmented architecture in which independently controllable coil zones are selectively energized based on vehicle position, enabling spatially selective power transfer and electronic rather than mechanical alignment.
- Developed a control concept -- Detect Estimate velocity Predict position Preselect coil Transfer Handover -- for predictive dynamic charging and seamless energy handover between zones.
- Proposed integrated sensing/health-monitoring layers using electrical and thermal signatures, with future ML-based classification for foreign-object detection and coil-health prediction explicitly treated as research directions.

Automated Electronics Disassembly System

- Developed a brand-agnostic mobile-phone depopulation machine for automated material recovery, connecting automation, electronics disassembly and sustainable materials recovery.

Sputtered RRAM Device Development

- Built a Cu/HfOx/ITO RRAM stack using RF/reactive sputtering and evaluated switching characteristics for emerging-memory and neuromorphic-computing applications.

ACHIEVEMENTS

- Foundation for Excellence Scholarship (FFE)- 2023
- SUN-FI Alumni Scholarship (2024)
- Startup Grant 50,000 - IIT Hyderabad BUILD Cohort (2025)
- Best Poster Award - Australia-India Critical Minerals Research Hub (2025)
- 2nd Runner-up - Inter-IIT Research Innovation in Science and Engineering (RISE) Competition (Grant Prize 2 Lakhs)
- Best UG Researcher of the Year Award @ IITH (2026)
- Manod Foundation Student Founder Fellowship(SFF) (2026)
- Selected for MIRAI-SETU Japan exchange program (2026)
- Selected for India Semiconductor Workforce Development Program (ISWDP) (2026)

SKILLS

Programming & Data: Python, C++, C, MATLAB, Minitab, Origin

AI / Data Science: Machine Learning, Deep Learning, statistical analysis, anomaly detection, process monitoring, AI-assisted workflow automation

Product / Problem Solving: Problem framing, experimentation, prototyping, product validation, root-cause analysis, data-driven decision making, technical communication

Semiconductor / Materials: Thin-film deposition, RF/DC sputtering, CVD, photolithography, thermal/e-beam evaporation, GI-XRD, SEM, SMU, vacuum systems, device modelling

Research Methods: Experimental design, SPC, failure analysis, materials characterization, process optimization, cross-domain research

PUBLICATIONS

- **Indian Patent No: 591727 "Sustainable Method for Delamination of Printed Circuit Boards to Recover Valuable Materials"**

CERTIFICATIONS

- Certified for the CREST fortnight residential summer school by CREST BITS Pilani and BITS Pilani, Hyderabad Campus, in collaboration with the National Skill Development Corporation for fabrication.
- Secured more than 75% in the Workshop titled CeNSE Winter School on Semiconductor Technology, conducted by CeNSE, IISC.
- Certified for the Statistical Process Control (SPC) bootcamp from Udemy.